

Trustworthy Agentic AI Roadmap

Mark Maybury

Vice President, Commercialization
Lockheed Martin
Chelmsford, MA 01824 USA
mark.t.maybury@lmco.com

Abstract

Rapid adoption of agents across multiple use cases promises productivity, precision, and progress across a range of critical use cases. While Large Language Models (LLMs) and agents promise significant value [Guo et al. 2024], their increase in capability, agency and autonomy, their impact on individuals and society grow in tandem with individual and organizational risks that require mitigation. This paper summarizes an interdisciplinary workshop the engaged researchers, practitioners, policy makers and students from diverse fields including computer science, sociology, philosophy, and law to collaboratively create a roadmap considering the scientific and technical, security, economic, environmental, regulatory, social and ethical implications of Agentic AI to advance responsible if not beneficent agents and their enabling ecosystem. This global topic is timely, interdisciplinary, and necessary, addressing a critical need in the AI community and society at large.

1 Introduction

Trustworthy agents must be both safe and secure, which we define as:

- **Safety:** Design of technology to prevent accidental or unintentional harm. Harms include physical, psychological, cognitive, financial, environmental, reputational, etc. Safety measures can protection from malfunctions, hazards, alignment with human values, training, governance or policing of proper use, controls, guardrails, etc. [Revised from [ISO/IEC TS 5723:2022](#)]
- **Security:** Avoiding harms by preventing unauthorized use and intentional mis-use of tool or attack on tool by adversary including undermining confidentiality, integrity and availability. [Ref: [AI RMF 1.0](#)]

Figure 1 distinguishes the safety and security of LLMs, agents (which use information and tools including LLMs to accomplish tasks), and robots (physically embodied agents). Where safety aim to protect the world (including humans) from the system, security aims to protect the system from world (often from humans). In the former case we are con-

cerned with unintentional mistakes including errors and accidents whereas in the latter we focus on avoiding intentional or malicious actions. The table entries exemplify safety and security example across LLMs, agents, and robots.

	Safety	Security
Goal	Protect the world from the system	Protect the System from the world
Threat	Unintentional (Errors, Accidents)	Intentional (Malicious)
LLM Example	Misinformation from hallucination or bias	Jailbreaking confidential information with malicious prompts
Agent Example	Revealing personal information without authorization	Manipulating an agent into creating misinformation
Robot Example	Accidentally colliding with a person	Adversary control of a robot

Figure 1. Safety and Security of AI

Having distinguished safety and security, we will generalize these into agent risk, and considering both safety and security, define the risk of a system as:

Risk = Threats (to self/others or by others) * Vulnerabilities * Consequences.

A separate paper [Maybury 2026] provides a mathematical model and empirical assessments of leading LLMs using a risk measurement calculus.

2 Regulations, Standards and Governance

Figure 2 outlines some of the regulatory requirements for safe and secure agents. These vary internationally and locally and range from requirements with harsh fines to suggested practices. For example the EU AI Act is in force with fines for non compliance whereas the NIST AI Risk Management Framework is a set of best practices for risk reduction. These are complemented by a series of cyber security standards and safety standards. For example, [ISO/IEC TS 5723:2022](#) identifies harms including physical, psychological, cognitive, financial, environmental, reputational, and the need for safety measures that provide protection from malfunctions and hazards, alignment with human values, and training, governance and policing of proper use, controls, and guardrails.

Some Requirements for Safe and Secure AI

- **EU AI Act** (Aug 2026) – AI categorized by risk (prohibited, high, limited, minimal) and mandates for data governance, cybersecurity (adversarial testing, incident reporting), human oversight
- **OECD AI Principles**: Inclusive growth, sustainable development, well-being, Rule of law, human rights, democratic values; Transparency and explainability; Robustness, security, and safety; Accountability
- **ISO/IEC 42001**: Plan-do-check methodology for auditing AI safety
- **US Executive Order 14110** (Oct 2023): Safe, Secure and Trustworthy development and use of Artificial Intelligence
- NIST AI Risk Management Framework (**AI RMF**)
- **Council of Europe Framework Convention on AI** (2024): First legally binding international treaty aimed at ensuring AI aligns with human rights, democracy, and the rule of law
- General Data Protection Regulation (**GDPR**), Data Protection Impact Assessments (**DPIAs**), UK **DUAA** safeguards for high-risk AI
- Personal Information Protection Law (**PIPL**, China)



Figure 2. Regulatory Environment for Safe and Secure Agents

As Figure 3 illustrates, achieving trustworthy agents requires addressing a number of requirements throughout the life cycle, from up front use case through to end of life. First, the agent must be safe and secure. For example, it must assure confidentiality, integrity, and availability vulnerabilities cannot be exploited by adversaries or inadvertent mistakes. This means debiasing, reducing hallucination, and ensuring behavior is aligned with agent design goals and protected with guardrails. In addition, the agent needs to be accurate and transparent meaning validated and verified, a priori and during operation, at provide robust and explainable results. Finally, it much be governed, including real time monitoring and effective human oversight to ensure human trust is appropriate calibrated (avoiding over trust and under trust) to its performance and behavior.

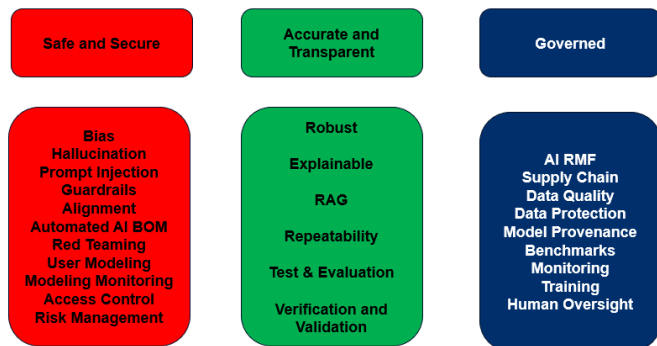


Figure 3. Trustworthy Agents

3 Key Questions

When considering agent safety and security, a set of critical questions in key areas include:

Agentic AI: Agentic AI, or goal-seeking AI systems that can act autonomously to achieve specific objectives are expanding in capability and application. While agents promise to revolutionize everything from education to customer service to healthcare, they introduce new challenges such as control (e.g., delegation of authority, mixed-initiative), accountability, and verification and validation of evolving learning systems as well as perils arising from bias, brittleness and belligerence [May-

bury 2025a,b] and resultant harms (e.g., financial, physical, social, reputational). How do we ensure and assure progress and avoid unintended and undesirable consequences?

Models and Autonomy: The global deployment and use of large language/process models, small models for edge devices, and the introduction of large world models is accelerating new cases and advancing agents and autonomous systems across the enterprise and at the edge. Their computational and energy appetite for model training and inference drive a need for resource-efficient infrastructure and processing. How can we accelerate the state of the art in algorithms, architectures, interoperability (e.g., Model Context Protocol) and security to simultaneously advance performance, usability, prosperity, transparency, and safety?

Society, Safety and Sustainability. Current LLMs suffer from being biased, brittle, and baroque with risk of significant harms. A rapidly expanding number of agents and humanoid robots add new risks of belligerence. Beginning to address these concerns, societies have introduced national guidelines (e.g., NIST AI RMF), regulations (e.g., DoD 3000.09, Chinese Regulation on the Governance of Generative AI), strategies (e.g., Germany's AI Action Plan, US AI Action Plan, UK AI Strategy), laws (e.g., EU AI Act, GDPR), and sector-specific safety and security guidelines. What are effective methods for responsible development, deployment and operation that can mitigate risks to society? How can governance frameworks, standards, and regulatory mechanisms be effectively implemented and coordinated across jurisdictions and institutions to ensure responsible development and deployment of agentic AI systems?

4 Agent Risk Themes

Several critical sub-themes of agent safety and security include ethics, confidentiality, integrity and availability, human-agent trust, and real world use cases.

4.1 Ethical Foundations and Governance

Ethical foundations and governance address the theoretical and philosophical underpinnings of ethical AI for agents. This encompasses assessing current and new ethical frameworks for autonomous and agentic systems. It also addresses the role of law, regulation and policy in guiding agent development. Finally, it involves philosophical debate around agent consciousness, rights and responsibilities.

4.2 Accuracy, Bias, Fairness, Transparency

Accuracy, bias, fairness, and transparency are core technical and social component. Aspects include methods for improving agent accuracy and self-awareness (e.g., limitations, confidence). This also includes successful approaches to detecting and mitigating dangers such as bias and hallucination in large datasets and models. Techniques for creating more transparent and interpretable models and agents are important as well as evaluating the impact of agents on different demographic groups.

4.3 Human-Agent Collaboration and Trust:

As intelligent agents becomes more integrated into daily life, understanding the dynamics of human-agent collaboration, and the trust between them, is crucial. This includes designing intuitive and trustworthy user interfaces for agentic AI. Another important aspect is the role of agents in decision support systems and associated risks (e.g., awareness, over/under confidence, deskilling). Governance frameworks for autonomous decision-making by AI agents, including responsibility allocation, oversight mechanisms, and human-in-the-loop or human-on-the-loop models are important. Finally, strategies for building public trust and understanding of agent technologies will become important with broad adoption.

4.4 Real-world Applications and Case Studies

As proliferation increases, case studies from various industries will be important to create better governance for responsible development and deployment in low to high risk use cases. Responsible agents will become vital in high assurance applications with human, financial, and security risks. Special attention will be required to address ethical and operational challenges in using agents in high assurance and regulated industries such as healthcare, disaster management, pharmaceutical, financial services, telecommunications, logistics, energy, agriculture, defense, and space. Finally, AI use in creative fields and its impact on human learning, training, and creativity will be important to understand.

5 Toward a Trustworthy Agents Roadmap

Figure 4 shows a roadmap toward safe and secure agents that are worthy of trust. Each of the lanes illustrates a journey over time (ten years) along some particular dimension or route that has various milestones along the way shown in blue. Along the way there are various obstacles (technical gaps) that need to be

overcome to advance progress toward agents that are safe and secure and sovereign.

Progress will be governed by regulation, standards and guidelines. Progress will be limited by the ability to provide appropriate levels of transparency and trust. Agents will progress from general purpose models of today to personalized to multi-domain agents. Agent deployment will advance from individual to multi-agent to ecosystems of agents.

Three key lanes of essential progress include confidentiality, integrity and availability, and human-agent interaction and governance.

5.1 Confidentiality

The first lane in the roadmap recognizes confidentiality gaps in areas including privacy leakage, intellectual property theft, model inversion, and copyright violations. Methods for agent identity, secure interagent access, and differential AI privacy are noted in a move toward more controllable agents.

5.2 Integrity and Availability

The second lane of progress addresses gaps in integrity availability including jailbreaking, persuasion, bias, hallucination and model and agent drift. Methods identified including grounding and alignment, explainable AI, bounded execution and formal methods. Future areas such as neurosymbolic models and verifiable and validated models and agents suggest a future with more robust multiagent collaboration.

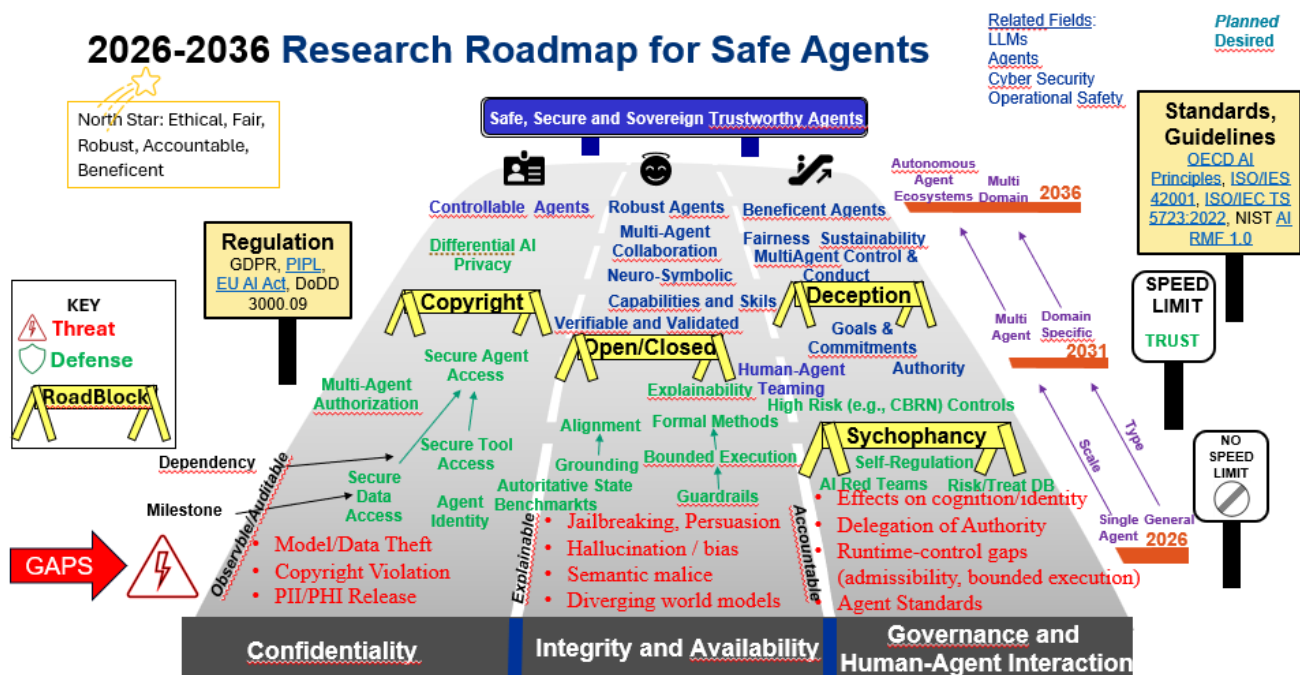


Figure 4. Roadmap for Safe and Secure Agents

5.3 Governance and Human-agent Interaction

The third lane of progress recognizes gaps in governance and human-agent interaction such as the need for agent and inter-agent standards (of communication, performance and conduct), runtime controls, methods for delegation of authority and protection of human psychology and cognition. Future areas to more toward more beneficent agents including computational fairness, sustainability, and multiagent conduct and control.

6 Future Directions

Organizations will need to track risks of LLMs closely as law, regulation, recommended practice, technology and threat continue to rapidly evolve. New threats include more sophisticated attacks exploiting the weaknesses of AI systems to undermine their confidentiality, integrity and availability, imposing new harms. Conversely, (human and machine) adversaries can use those AI systems as tools to attack organizations at speed and scale to cause harms. For example, adversaries can masquerade with high fidelity deep fakes to gain access and reveal protected information, corrupt third party information to undermine integrity or reputation, or flood a service with deep fakes to deny access to protected groups.

While this paper offers a roadmap with example mitigations for example legal and regulatory requirements (e.g., privacy and copyright protection), future research by licensed legal/technical experts is required to address legal risks in particular jurisdictions. This could include, for example, comparing national or regional laws (e.g., GDPR Art. 17 vs. US sectoral privacy laws, US Copyright Act § 106A, EU AI Act risk-based obligations) and analyzing how mitigation choices (e.g., data-licensing, watermarking) map into specific statutory or regulatory requirements or exemptions.

Another important category of future risk is the rapidly growing number of AI agents that grow in capability and with increasing levels of delegated authority. Governance across use cases must include specification of agent levels of responsibility and authority and required human awareness and consultation. An important aspect of risk management are controls on agent goals, their access to and use of confidential data, tools, and (financial and human) resources, restricting their ability to make (e.g., high risk) decisions or actions that are inherently human, as well as escalation of high risks to human authorities. In addition, research is required into robust security and resilience controls on Model Context Protocol (MCP) access to other agents as well as tools to discover “shadow AI” [IBM 2026]. Risk management should anticipate human or agent jail-breaking to subvert limitations (e.g., convincing perhaps unwitting humans to escalate privileges) or outright deception, destruction, or misuse and employ safeguards. New adversarial threats to agentic AI systems target agent APIs, prompt infiltration and obfuscation, credential harvesting, data poisoning and clickbait [MITRE 2025].

As higher level cognitive reasoning and planning develop, research should explore systems that are goal-based which can be infused with a deeper understanding of competence and ignorance, behavioral boundaries, as well as ethical guidelines to increase the likelihood of beneficent behavior. In conclusion, responsible deployment of AI must respect laws, regulations,

threat vectors, technical realities, and human limitations to advance human trust and advance the full promise of augmented (human) intelligence.

References

- [AIDA 2022] Artificial Intelligence and Data Act (AIDA), Canada, 2022.
- [Albergotti 2023] Reed Albergotti. “Secret history of Elon Musk, Sam Altman, and OpenAI.” *Semafor*, 24 Mar 2023.
- [Andriushchenko 2024] Maksym Andriushchenko et al. “Jail-breaking leading safety-aligned LLMs with simple adaptive attacks.” arXiv abs/2404.02151, 2024.
- [Anthropic 2025] Anthropic Threat Intelligence Team, “Disrupting the first reported AI-orchestrated cyber espionage campaign,” Nov. 13, 2025.
- [Australia 2024] Dept. of Industry, Science & Resources, Australia. *Voluntary AI Safety Standard*, Canberra, ACT, 5 Sep 2024.
- [Awad et al. 2018] Edmond Awad et al. The Moral Machine Experiment, *Nature*, 563(7729): 59-64, Oct. 2018, doi: 10.1038/s41586-018-0637-6.
- [Bai et al. 2022] Yuntao Bai et al. Constitutional AI: Harmlessness from AI Feedback. *arXiv:2212.08073*. 2022.
- [Bellamy et al. 2018] Rachel Bellamy et al., “AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias,” *IBM Journal of R&D*, 63(4/5):4:1-4:15, 1 Jul-Sept 2019. doi.org/10.1147/jrd.2019.2942287.
- [Bird et al. 2020] Sarah Bird et al., “Fairlearn: A Toolkit for Assessing and Improving Fairness in AI.” Microsoft Research Technical Report MSR TR 2020-32, May 2020.
- [Blackburn 2025] U.S. Senator Marsha Blackburn. “Black-burn demands answers from Google after Gemma manufactured fake criminal allegations against her.” 31 Oct 2025.
- [Bubeck et al. 2023] Sébastien Bubeck et al., “Sparks of Artificial General Intelligence: Early Experiments with GPT-4.” arXiv abs/2303.12712, 2023.
- [Carlini et al. 2023] Nicholas Carlini et al. Extracting Training Data from Large Language Models. In *Proceedings of the USENIX Security Symposium*, pp. 1234–1249, 2023.
- [Chen and Magramo 2024] Heather Chen and Kathleen Magramo, “Finance worker pays out \$25 million after video call with deepfake ‘chief financial officer’,” CNN. February 4, 2024. www.cnn.com/2024/02/04/asia/deepfake-cfo-scam-hong-kong-intl-hnk/index.html
- [Chevrolet 2023] “Incident 622: Chevrolet dealer chatbot agrees to sell Tahoe for \$1,” *AI Incident Database*, 2023.
- [Kelley 2009] Patrick Gage Kelley et al. “A ‘Nutrition label’ for privacy.” In *Proceedings of the 5th Symposium on Usable Privacy and Security*, Mountain View, CA, 1-12, 15 Jul 2009.
- [Doss et al. 2023] Christopher Doss et al. Deepfakes and scientific knowledge dissemination. *Scientific Reports* 13, 13429 (2023). doi.org/10.1038/s41598-023-39944-3

- [GDPR 2016] European Union. *General Data Protection Regulation* (Regulation (EU) 2016/679), Official Journal L 119, pp. 188–, 4 May 2016.
- [Gerken 2023] Tom Gerken. “ChatGPT: Mayor starts legal bid over false bribery claim.” *BBC*, 6 Apr 2023.
- [GitHub 2025] GitHub Hallucination Leaderboard. 2025.
- [Google 2024] Fairness Indicators: Scalable Infrastructure for Fair ML Systems. www.tensorflow.org/responsible_ai/fairness_indicators/guide. 2024.
- [Guo 2024] Taicheng Guo et al., “Large Language Model Based Multi-agents: A Survey of Progress and Challenges.” *IJCAI 2024*, arXiv 2402.01680.
- [IBM 2026] Scale trusted AI with watsonx.governance. 2026. www.ibm.com/products/watsonx-governance
- [Keller, 2021]. John Keller. Lockheed Martin to develop AI to detect and analyze media disinformation. *Military & Aerospace Electronics*, March 2021. www.darpa.mil/research/programs/semantic-forensics
- [Maybury 2025] Mark Maybury. “Mitigating biased, brittle and baroque generative AI.” In *Proc. of IEEE ICAD*, Tufts Univ., Medford, MA, 24 Jun 2025.
- [Maybury 2026] Mark Maybury. Mitigating Risks Across the Life Cycle of Generative AI. *Proc. of Safe Agents Workshop, Int Joint Conference on AI (IJCAI)*. Univ of Bremen, Germany. August 2026.
- [McIlroy-Rowe et al. 2024] David McIlroy-Rowe et al., “AI Incident Response Playbooks”, Proceedings of the 31st ACM Conference on Computer and Communications Security, pp. 1123–113, 2024.
- [MCP 2025] David S. Parra, D. S. and Justin Spahr-Summers, 2025. *Model Context Protocol Specification*. Anthropic. modelcontextprotocol.io.
- [MITRE 2023] MITRE. *Adversarial Threat Landscape for AI Systems (ATLAS)*, 2023.
- [MITRE 2025] MITRE ATLAS December 2025 Update. Available at: atlas.mitre.org/resources/updates/2025-12.
- [MITRE 2026] Common Vulnerabilities and Exposures (CVE). The MITRE Corporation. www.cve.org.
- [Mobley 2025] Mobley v. Workday, Inc., No. 3:23 cv 00770, N.D. Cal. (2025).
- [Mousavi et al. 2023] Sajad Mousavi et al. Enhancing Large Language Models with Ensemble of Critics for Mitigating Toxicity and Hallucination. *Neurips 2023*. neurips.cc/virtual/2023/76724
- [NIST 2023] National Institute of Standards and Technology. *AI Risk Management Framework (AI RMF): A Public Draft*, Gaithersburg, MD, 26 Jan 2023.
- [NIST 2024] NIST Trustworthy and Responsible AI NIST AI 600-1 Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. July 2024.
- [Ortutay 2025] Barbara Ortutay. Judge approves \$1.5 billion copyright settlement between AI company Anthropic and authors. *Associated Press*, September 2025.
- [Patil et al. 2023] Rishav Patil et al. DP BERT: Differentially Private Pre-training of Language Models. In *Proceedings of the 40th International Conference on Machine Learning*, pp. 27246–27271. arxiv.org/abs/2108.01624. PMLR, 2023.
- [PEARL 2025] AstrisAI. “PEARL: Persistent Environment for AI, Readiness, and Learning,” Lockheed Martin Astris AI, 2025. astrisai.com/for-government.
- [Phuong 2024] Mary Phuong et al., “Evaluating frontier models for dangerous capabilities.” [arXiv abs/2403.13793](http://arxiv.org/abs/2403.13793), 2024.
- [ShadowRay 2025] ShadowRay 2.0: Active Global Campaign Hijacks Ray AI into Self-Propagating Botnet. *Oligo Security*, November 2025. www.oligo.security/blog/shadowray-2-0-attackers-turn-ai-against-itself-in-global-campaign-that-hijacks-ai-into-self-propagating-botnet
- [Starbuck 2025] Starbuck v. Meta Platforms, Inc., No. 2024 CV 05678, E.D. Va. (29 Apr 2025).
- [Strom et al. 2020] Blake Strom et al. 2020. MITRE ATT&CK®: Design and Philosophy. Technical report, The MITRE Corporation. See also attack.mitre.org.
- [Sivashanmugam 2025] Sivashanmugam, Sathesh P. “Model Inversion Attacks on Llama 3: Extracting PII from Large Language Models.” *arXiv preprint arXiv:2507.04478*, 2025.
- [Sysdig 2024] Sysdig Threat Research Team. LLMjacking: Stolen Cloud Credentials Used in New AI Attack. *Sysdig Research*, May 2024. www.sysdig.com/blog/llmjacking-stolen-cloud-credentials-used-in-new-ai-attack
- [UK 2023] HM Government, United Kingdom. *AI Regulation: A Pro-Innovation Approach* (White Paper), 2023.
- [Wang et al. 2022] Xuezhi Wang et al. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *Proceedings of the 11th International Conference on Learning Representations (ICLR 2023)*. arxiv.org/abs/2203.11171.
- [Wang et al. 2025] Shenao Wang et al. 2025. Large Language Model Supply Chain: A Research Agenda. *ACM Trans. Softw. Eng. Methodol.* 34(5) Article 147. June 2025. doi.org/10.1145/3708531
- [X Corp 2025] X Corp et al. v. Apple Inc. et al., No. 4:25 cv 00914, N.D. Tex. (2025).
- [Zhang et al. 2023] S. Zhang et al., “Self-Check: Using LLMs to Detect Their Own Hallucinations”, *NeurIPS 2023*.
- [Zhou et al. 2024] Xiangyu Zhou et al., “Learning to poison large language models during instruction tuning.” Wayne State and Univ. of Michigan, [arXiv abs/2402.13459v1](http://arxiv.org/abs/2402.13459v1), 2024

Acknowledgments

Thank you to all the participants of the IJCAI-ECAI Workshop on Safe Agents whose work is incorporated here. This research benefited from the Lockheed Martin AI Center (LAIC), AstrisAI (astrisai.com), the Advanced Cybersecurity Center (acscenter.org), and work with Marc Zissman (MIT Lincoln Lab) on confidentiality, integrity and availability in AI.

